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Questions

Q1.

Which statement is correct?

(1)

- ☒ **A** barium carbonate is less stable to heat than magnesium carbonate
- ☒ **B** barium hydroxide is less soluble in water than magnesium hydroxide
- ☒ **C** barium sulfate is less soluble in water than magnesium sulfate
- ☒ **D** barium metal is less reactive with water than magnesium metal

(Total for question = 1 mark)

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Q2.

This question is about compounds containing Group 2 elements.

(a) Hydrated magnesium nitrate has the formula $Mg(NO_3)_2.xH_2O$.

A student devised an experiment to determine the value of **x** by leaving 5.12 g of hydrated magnesium nitrate for several hours in a warm oven. After this time, the solid remaining had a mass of 2.97 g.

(i) State why the student used a warm oven to remove the water from the hydrated salt, rather than direct heating with a Bunsen burner.

(1)

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(ii) Use the data obtained by the student to calculate the value of **x**.
You **must** show your working.

(4)

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(b) Magnesium carbonate, MgCO3, is a white solid used as an additive in foods.

(i) Explain, in terms of energy changes, why magnesium carbonate is insoluble in water.

(2)

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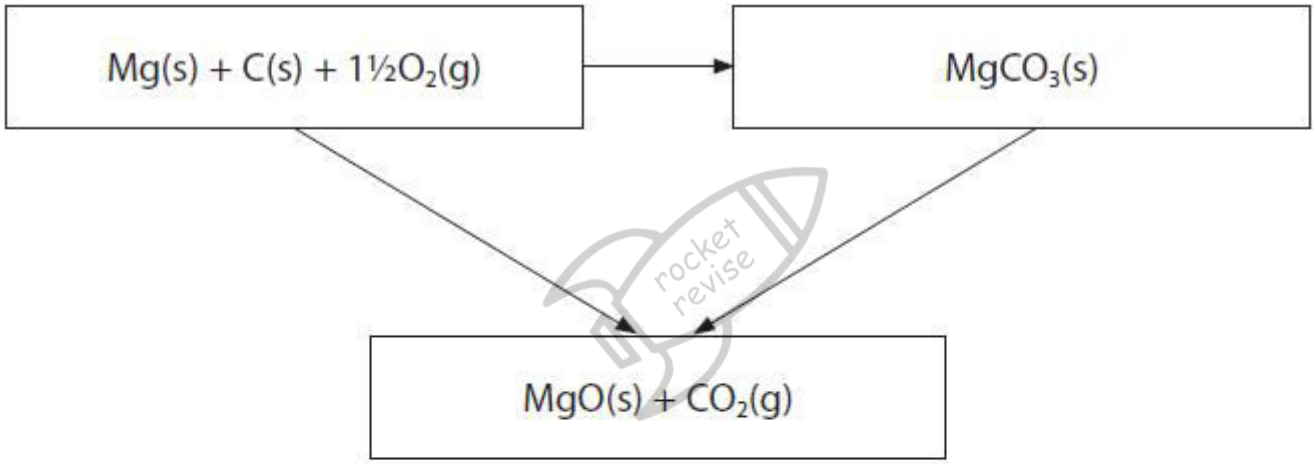
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(ii) The Hess cycle and data to calculate the enthalpy change for the thermal decomposition of MgCO3 are shown.

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Compound	$\Delta_f H^{\ominus}$ / kJ mol^{-1}
$CO_2(g)$	−394
MgO(s)	−602
$MgCO_3(s)$	−1096

Calculate the enthalpy change for the thermal decomposition of $MgCO_3$.

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(2)

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(iii) Explain the trend in thermal stability of Group 2 carbonates.

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(Total for question = 13 marks)

Q3.

Equations for four reactions of copper or its compounds are shown.

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Reaction 1 $2Cu^{2+} + 4I^- \rightarrow 2CuI + I_2$

Reaction 2 $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

Reaction 3 $2Cu^+ \rightarrow Cu + Cu^{2+}$

Reaction 4 $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

(a) Which is a disproportionation reaction?

(1)

- ☒ **A** Reaction 1
- ☒ **B** Reaction 2
- ☒ **C** Reaction 3
- ☒ **D** Reaction 4

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(b) Which is an acid-base reaction?

(1)

- ☐ **A** Reaction 1
- ☐ **B** Reaction 2
- ☐ **C** Reaction 3
- ☐ **D** Reaction 4

(Total for question = 2 marks)

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Q4.

Group 2 hydroxides, $M(OH)_2$, are used to neutralise acids.

(a) Write an equation for the reaction of calcium with cold water.

State symbols are not required.

(1)

(b) Explain why a saturated solution of calcium hydroxide is more alkaline than a saturated solution of magnesium hydroxide.

(2)

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(c) A solution was tested for the presence of carbonate ions.

Dilute hydrochloric acid was added to the solution and the gas given off was bubbled through limewater (aqueous calcium hydroxide). A white precipitate formed.

(i) Write the **ionic** equation for the formation of the gas.
State symbols are not required.

(1)

(ii) Write the equation for the formation of the precipitate.
Include state symbols.

(2)

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(d) Magnesium hydroxide can be used to neutralise hydrochloric acid in the stomach to relieve acid indigestion.

$$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$$

Calculate the minimum mass of magnesium hydroxide needed to neutralise 0.150 mol of hydrochloric acid.
Give your answer to an appropriate number of significant figures.

(3)

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(Total for question = 9 marks)

Q5.

Exactly 50.0 cm^3 of 2.00 mol dm^{-3} nitric acid reacts with 50.0 cm^3 of 1.00 mol dm^{-3} barium hydroxide to form a neutral solution of barium nitrate.

(a) What is the concentration, in mol dm^{-3}, of barium nitrate in the solution?

(1)

- ☒ **A** 0.05
- ☒ **B** 0.50
- ☒ **C** 1.00
- ☒ **D** 2.00

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(b) The volume of the nitric acid is measured using a burette.

Each burette reading has an uncertainty of ± 0.05 cm^3.
What is the percentage uncertainty in measuring 50.00 cm^3 of the nitric acid?

(1)

- ☒ **A** ± 0.40%
- ☒ **B** ± 0.20%
- ☒ **C** ± 0.10%
- ☒ **D** ± 0.05%

(Total for question = 2 marks)

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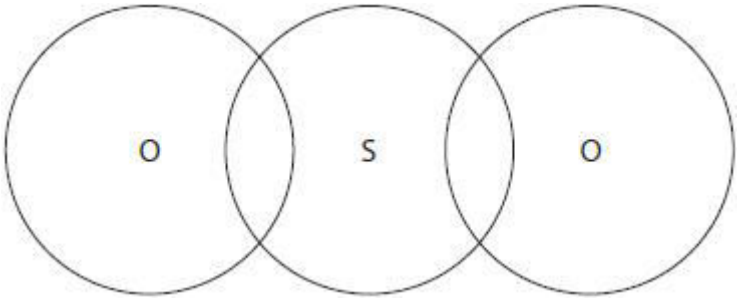
Q6.

Major volcanic eruptions, such as the one in Indonesia in 1815, eject large amounts of ash and gases into the atmosphere. The gases include carbon dioxide, water vapour, hydrogen chloride and sulfur dioxide.

(a) (i) Complete the dot-and-cross diagram to show a possible arrangement of the outer shell electrons in a molecule of sulfur dioxide.

Use dots (•) for the sulfur electrons and crosses (×) for the oxygen electrons.

(2)



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(ii) Suggest a value for the bond angle.

(1)

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(b) The levels of sulfur dioxide in the atmosphere can be measured by collection and titration with sodium hydroxide solution.

One possible method is:

- air containing sulfur dioxide is bubbled through hydrogen peroxide solution
- all the sulfur dioxide reacts to form 40.0 cm^3 of dilute sulfuric acid, solution Z

$$SO_2(g) + H_2O_2(aq) \rightarrow H_2SO_4(aq)$$

- a pipette is used to remove 10.0 cm^3 portions of solution Z
- each portion is titrated with 0.00500 mol dm^{-3} sodium hydroxide.

The results are shown in the table.

Titration	1	2	3
Final volume / cm^3	21.55	42.70	21.05
Initial volume / cm^3	0.00	21.55	0.00
Titre / cm^3	21.55	21.15	21.05

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The mean titre is 21.10 cm^3

(i) Give a reason why a further titration was not attempted.

(1)

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(ii) Calculate the number of moles of sulfuric acid in the 40 cm^3 of solution Z.

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(aq)$$

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(3)

(iii) Deduce the number of moles of sulfur dioxide bubbled through the hydrogen peroxide solution, using the answer from (b)(ii) and the equation

$$SO_2(g) + H_2O_2(aq) \rightarrow H_2SO_4(aq)$$

(1)

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(iv) The air containing sulfur dioxide was bubbled through the hydrogen peroxide solution at a rate of 10 $dm^3\ min^{-1}$ for 30 minutes.
Calculate the concentration of sulfur dioxide in the air in parts per million (ppm) by volume.
The molar volume of a gas at r.t.p. is 24 $dm^3\ mol^{-1}$.

(3)

(c) (i) During a volcanic eruption, hydrogen chloride gas is also released into the upper atmosphere, which in turn produces some chlorine free radicals. Chlorine free radicals react with ozone:

$$Cl^{\bullet} + O_3 \rightarrow ClO^{\bullet} + O_2$$

$$ClO^{\bullet} + O_3 \rightarrow Cl^{\bullet} + 2O_2$$

Derive the overall equation for this reaction of ozone.
State symbols are not required.

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(1)

(ii) Give **two** reasons why the presence of a small number of chlorine free radicals in the upper atmosphere causes a large decrease in the amount of ozone.

(2)

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(d) (i) Sulfur dioxide is converted into sulfur trioxide in the upper atmosphere in the presence of ultraviolet light.

$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

Show, by use of all the relevant oxidation numbers, that this is a redox reaction.

(2)

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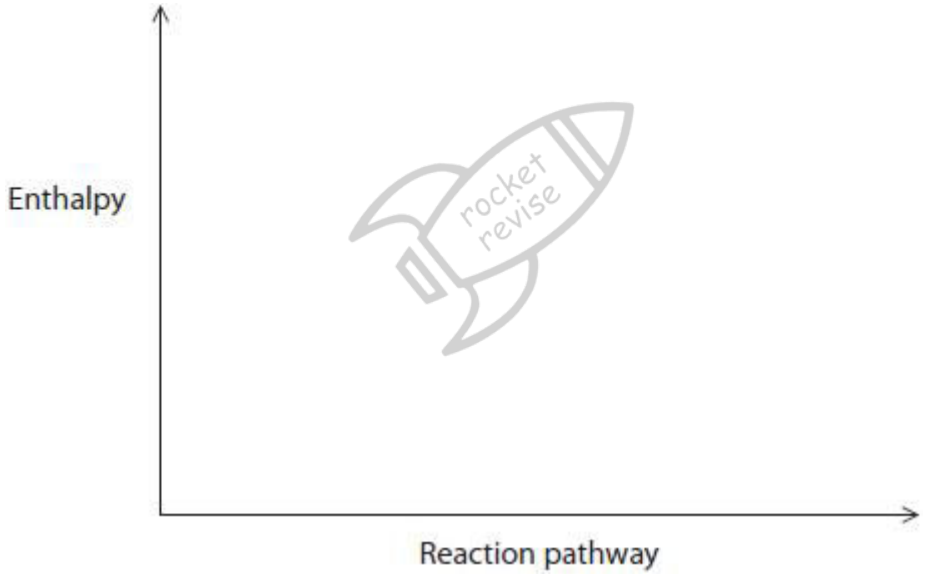
(ii) Sulfur trioxide reacts with water to produce sulfuric acid.

$$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq) \qquad \Delta_r H = -200 \text{ kJ mol}^{-1}$$

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Draw an enthalpy level diagram to show the enthalpy change for this reaction.

(2)



(iii) Droplets of sulfuric acid are formed when sulfur trioxide reacts with water in the upper atmosphere. These droplets reflect ultraviolet radiation from the Sun back into space.

In the volcanic eruption in Indonesia in 1815, large amounts of both carbon dioxide and sulfur dioxide were released.

During the following year, there were global low temperatures.

Assess the effect of **both** gases on global temperatures after the volcanic eruption.

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(Total for question = 21 marks)

Q7.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these sulfates is the **least** soluble in water?

- ☒ **A** $CaSO_4$
- ☒ **B** $BaSO_4$
- ☒ **C** K_2SO_4
- ☒ **D** Rb_2SO_4

(Total for question = 1 mark)

Q8.

Limewater is a solution of calcium hydroxide used in the laboratory to test for carbon dioxide.

(a) Write the equation for the formation of the white precipitate in this test. Include state symbols.

(2)

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(b) The concentration of a saturated solution of calcium hydroxide can be determined by titration.

25.0 cm^3 portions of a saturated solution of calcium hydroxide were titrated with 0.0500 mol dm^{-3} hydrochloric acid
using phenolphthalein indicator. The mean titre was 23.40 cm^3.
Calculate the concentration of calcium hydroxide in g dm^{-3}.
Give your answer to an appropriate number of significant figures.
[$M_r(Ca(OH)_2) = 74.1$]

(4)

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(c) The experiment was repeated using the same hydrochloric acid with a saturated solution of magnesium hydroxide.

Explain the difference (if any) in the mean titre.

(2)

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(Total for question = 8 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Compound **Q** gives off nitrogen dioxide when heated, and produces a red colour in a flame test.

(a) Which of these compounds could be **Q**?

(1)

- ☐ **A** barium nitrate
- ☐ **B** lithium nitrate
- ☐ **C** magnesium nitrate
- ☐ **D** rubidium nitrate

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(b) Which colour and test results are correct for nitrogen dioxide gas?

(1)

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	Colour of nitrogen dioxide	Colour change of damp litmus paper
☒ **A**	brown	blue to red
☒ **B**	brown	red to blue
☒ **C**	colourless	blue to red
☒ **D**	colourless	red to blue

(Total for question = 2 marks)

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Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In an oxide of potassium, the oxidation number of oxygen is $-\frac{1}{2}$.

What is the formula of this oxide?

☒ **A** K_2O
☒ **B** K_2O_2
☒ **C** K_2O_3
☒ **D** KO_2

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(Total for question = 1 mark)

Q11.

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Magnesium oxalate, MgC_2O_4 , decomposes on heating to form magnesium carbonate and carbon monoxide.

$$MgC_2O_4(s) \rightarrow MgCO_3(s) + CO(g)$$

(a) A 6.0 g sample of magnesium oxalate was heated for three minutes but the decomposition was only 70 % complete.

Calculate the total mass of solid that remains.

(4)

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(b) In practice, magnesium carbonate also decomposes on stronger heating.

Describe and explain the trend in the thermal decomposition of Group 2 carbonates.

(3)

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(c) Suggest why a pure sample of magnesium carbonate will **not** be produced from the decomposition of magnesium oxalate even if the sample is heated for longer.

(1)

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(Total for question = 8 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) A pellet of sodium hydroxide has a mass of 0.700 g.

Some pellets were dissolved to make 350 cm^3 of 0.25 mol dm^{-3} solution.

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[M_r value: NaOH = 40]

How many pellets were dissolved?

(1)

- ☐ **A** 4
- ☐ **B** 5
- ☐ **C** 8
- ☐ **D** 125

(b) 25.0 cm^3 of the sodium hydroxide solution prepared in (a) was placed in a conical flask and titrated with sulfuric acid.

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

Calculate the number of moles of sulfuric acid that reacted.

(1)

- ☐ **A** 0.0031
- ☐ **B** 0.0063
- ☐ **C** 0.013
- ☐ **D** 0.044

(c) Phenolphthalein indicator was used for the titration in (b).

What was the colour change at the endpoint?

(1)

- ☐ **A** colourless → pink
- ☐ **B** pink → colourless
- ☐ **C** orange → yellow
- ☐ **D** yellow → orange

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(Total for question = 3 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Barium chloride solution, $BaCl_2(aq)$, reacts with gallium sulfate solution, $Ga_2(SO_4)_3(aq)$ to form a precipitate of barium sulfate, $BaSO_4(s)$.

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What is the minimum volume of 0.100 mol dm^{-3} barium chloride needed to precipitate **all** the sulfate ions in 200 cm^3 of 0.05 mol dm^{-3} gallium sulfate?

☐ **A** 100 cm^3

☐ **B** 200 cm^3

☐ **C** 300 cm^3

☐ **D** 400 cm^3

(Total for question = 1 mark)

Q14.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which of these compounds does **not** produce a colour in a flame test, and produces an alkaline gas when warmed with sodium hydroxide solution?

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☐ **A** $Ca(OH)_2$

☐ **B** $Mg(OH)_2$

☐ **C** NH_4Cl

☐ **D** $BeCl_2$

(Total for question = 1 mark)

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Q15.

This question is about ethanoic acid and some related salts.

(a) A test to confirm the presence of an aqueous acid is adding a small amount of solid sodium carbonate to the solution.

Describe **two** observations you would **see** in this test.

(2)

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(b) Sodium ethanoate is a component of reusable hand warmers. In use, a supersaturated solution of sodium ethanoate recrystallises to form solid hydrated sodium ethanoate, releasing energy.

$CH_3COONa(aq) + 3H_2O(l) \rightarrow CH_3COONa.3H_2O(s)$ $\Delta_r H = -19.7$ kJ mol^{-1}

A hand warmer has a mass of 63.2 g and forms 20.1 g of hydrated sodium ethanoate on recrystallisation.

Calculate the maximum temperature reached by the hand warmer if its initial temperature is 5.0°C.

[Specific heat capacity of the hand warmer = 3.0 J °C^{-1} g^{-1}]

(5)

(c) Ammonium ethanoate, $CH_3COONH_4(s)$, is used to control the pH of foods. It can be formed by the reaction of pure ethanoic acid, $CH_3COOH(l)$, with ammonium carbonate, $(NH_4)_2CO_3(s)$.

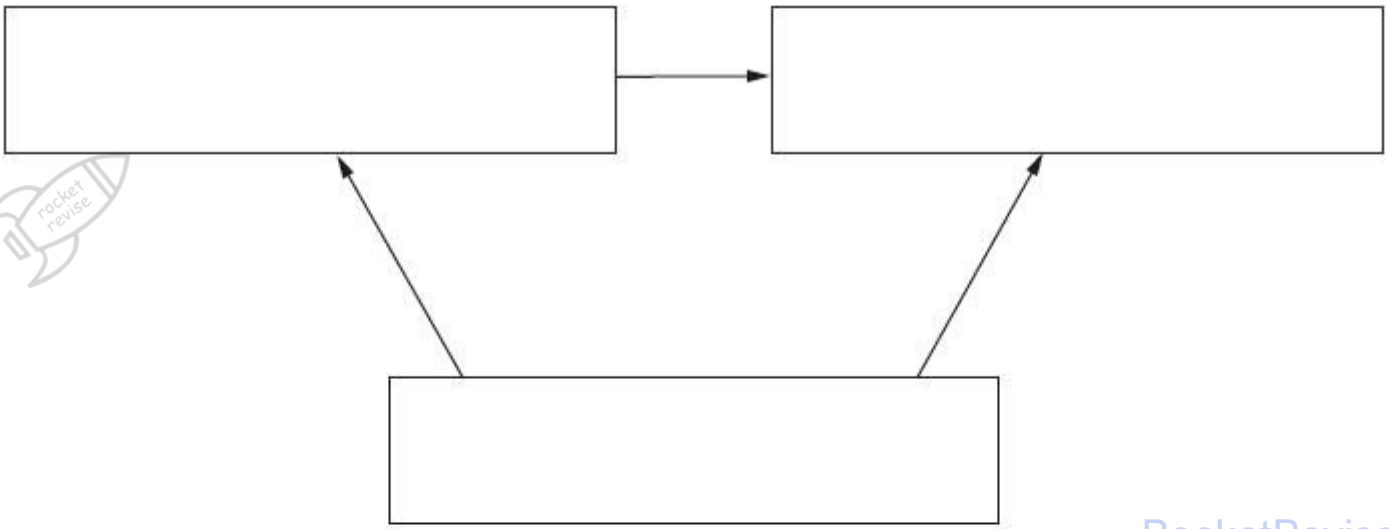
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Calculate the standard enthalpy change for this reaction by completing the Hess cycle and using the data shown.

(5)

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Compound	Enthalpy change of formation / kJ mol^{-1}
$CH_3COOH(l)$	−484.5
$(NH_4)_2CO_3(s)$	−939.9
$CH_3COONH_4(s)$	−586.3
$CO_2(g)$	−393.5
$H_2O(l)$	−285.8

(d) Ammonium carbonate, $(NH_4)_2CO_3$, is an ingredient in cleaning solutions for camera lenses.

These are aqueous solutions which contain no more than 1.8 g of ammonium carbonate in 100 cm^3 of solution.
Calculate the maximum concentration, in mol dm^{-3}, of ammonium carbonate in such a solution.

(2)

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(Total for question = 14 marks)

Q16.

This question is about Group 1 metals.

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(a) When potassium is placed into a beaker of cold water, potassium hydroxide and hydrogen are formed.

(i) Write the equation for this reaction. Include state symbols.

(2)

(ii) This is a redox reaction.
State which element is oxidised and which is reduced.
Justify your answer by giving the initial and final oxidation numbers of any element that changes oxidation state.

(2)

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(iii) The reaction of potassium with water is very vigorous and a flame is seen.
State the colour of the flame.

(1)

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(b) The label has come off a bottle known to contain **M**, a Group 1 metal which is stored in oil.

A student carried out an experiment to determine the identity of **M**.

Procedure

Step **1**

A small piece of **M** was wiped with tissue paper to remove the oil.
The piece of **M** was weighed and placed in a beaker of distilled water.

Step **2**

After the reaction had finished, the contents of the beaker and washings were transferred to a 250.0 cm^3 volumetric flask. The solution was made up to the mark with distilled water and mixed thoroughly.

Step **3**

A pipette was used to transfer 25.0 cm^3 portions of this solution to conical flasks. Each portion was then titrated with hydrochloric acid of concentration 0.400 mol dm^{-3}.

Results

Mass of metal, M	0.37 g
Mean titre of hydrochloric acid	12.80 cm^3

The reaction taking place is shown.

$$\mathbf{M}OH(aq) + HCl(aq) \rightarrow \mathbf{M}Cl(aq) + H_2O(l)$$

(i) The indicator used was phenolphthalein.

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State the colour **change** at the end-point.

(2)

from ..... to .....

(ii) Calculate the relative atomic mass of **M** and use it to identify the Group 1 metal, **M**.

(4)

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(c) Another student repeated the experiment, using a different sample of metal **M**, but did not wipe off the oil before weighing it.

State how this would change the calculated value of the relative atomic mass of **M**. Justify your answer.

(2)

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(Total for question = 13 marks)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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What volume, in dm^3, of hydrogen gas will be produced when 3.00 g of lithium is reacted with water at room temperature and pressure (r.t.p.)?

$$2Li + 2H_2O \rightarrow 2LiOH + H_2$$

[Molar volume of gas at r.t.p. = 24.0 dm^3 mol^{-1}]

- ☒ **A** 0.217
- ☒ **B** 0.435
- ☒ **C** 5.22
- ☒ **D** 10.4

(Total for question = 1 mark)

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Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Several factors may affect ionisation energies:

I) the number of protons increases

II) the outer electron is further from the nucleus

III) the amount of shielding increases

IV) the number of unpaired outer electrons increases

Which factors explain the **decrease** in ionisation energy as Group 1 is descended?

- ☒ **A** I and II
- ☒ **B** II and III
- ☒ **C** III and IV
- ☒ **D** I, II, III and IV

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(Total for question = 1 mark)

Q19.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Magnesium nitrate decomposes on heating.

$$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$$

What is the total volume of gas formed at room temperature and pressure (r.t.p.) when 0.005 mol of

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magnesium nitrate decomposes completely?

[Molar volume of a gas at r.t.p. = 24 000 cm^3 mol^{-1}]

☐ **A** 600 cm^3

☐ **B** 300 cm^3

☐ **C** 240 cm^3

☐ **D** 120 cm^3

(Total for question = 1 mark)

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Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A 4.00 mol dm^{-3} solution of an acid is used to prepare dilute solutions of the acid.

What volume of water is required to make up 150 cm^3 of 0.35 mol dm^{-3} solution of the acid?

☐ **A** 13.1 cm^3

☐ **B** 52.5 cm^3

☐ **C** 97.5 cm^3

☐ **D** 136.9 cm^3

(Total for question = 1 mark)

Q21.

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Data for some Group 1 and Group 2 cations are shown in the table.

Cation	Ionic radius / nm	Ionic charge
W	0.100	+2
X	0.138	+1
Y	0.113	+2
Z	0.149	+1

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Which cation would be expected to form the nitrate with the **greatest** thermal stability?

- ☒ **A** W
- ☒ **B** X
- ☒ **C** Y
- ☒ **D** Z

(Total for question = 1 mark)

Q22.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which property **decreases** as Group 2 of the Periodic Table is descended?

- ☒ **A** reactivity of the elements
- ☒ **B** solubility of the hydroxides
- ☒ **C** solubility of the sulfates
- ☒ **D** thermal stability of the carbonates

(Total for question = 1 mark)

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Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student carries out two sets of titrations, one using methyl orange and the other using phenolphthalein as indicators.

The conical flask contains sodium hydroxide solution and the burette contains hydrochloric acid.

What are the colour changes at the end-points?

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		Methyl orange	Phenolphthalein
☐	A	red to orange	colourless to pink
☐	B	yellow to orange	pink to colourless
☐	C	red to orange	pink to colourless
☐	D	yellow to orange	colourless to pink

(Total for question = 1 mark)

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Q24.

Magnesium ethanedioate (MgC_2O_4) decomposes on gentle heating to form magnesium carbonate and carbon monoxide.

$$MgC_2O_4 \rightarrow MgCO_3 + CO$$

(a) (i) State why the thermal decomposition of magnesium ethanedioate should be carried out in a fume cupboard.

(1)

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(ii) After heating a 6.0 g sample of magnesium ethanedioate for three minutes, the decomposition was 70 % complete.

Calculate the total mass of the solid mixture that remains.

(4)

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(b) Magnesium carbonate undergoes thermal decomposition at a higher temperature than magnesium ethanedioate.

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$MgCO_3 \rightarrow MgO + CO_2$

Explain the trend in the thermal decomposition of Group 2 carbonates going down the group.

(3)

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(Total for question = 8 marks)

Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is the ionic equation for the reaction of solid barium carbonate with dilute hydrochloric acid?

☐ **A** $BaCO_3(s) + 2H^+(aq) \rightarrow Ba^{2+}(aq) + CO_2(g) + H_2O(l)$

☐ **B** $Ba_2CO_3(s) + 2H^+(aq) \rightarrow 2Ba^+(aq) + CO_2(g) + H_2O(l)$

☐ **C** $CO_3^{2-}(s) + 2H^+(aq) \rightarrow CO_2(g) + H_2O(l)$

☐ **D** $CO_3^{2-}(s) + 2HCl(aq) \rightarrow 2Cl^-(aq) + CO_2(g) + H_2O(l)$

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(Total for question = 1 mark)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which nitrate forms oxygen as the **only** gaseous product on heating?

☐ **A** $LiNO_3$

☐ **B** $NaNO_3$

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☒ **C** $Mg(NO_3)_2$
☒ **D** $Ca(NO_3)_2$

(Total for question = 1 mark)

Q27.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A small piece of calcium is added to a beaker of distilled water.

(a) Which is the equation for this reaction?

(1)

☒ **A** $Ca + H_2O \rightarrow CaO + H_2$
☒ **B** $Ca + 2H_2O \rightarrow CaO_2 + 2H_2$
☒ **C** $Ca + H_2O \rightarrow CaOH + \frac{1}{2} H_2$
☒ **D** $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$

(b) Which element is oxidised and which is reduced when calcium reacts with water?

(1)

	Element oxidised	Element reduced
☒ **A**	calcium	hydrogen
☒ **B**	calcium	oxygen
☒ **C**	hydrogen	calcium
☒ **D**	hydrogen	oxygen

(Total for question = 2 marks)

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Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The presence of ammonium ions in a compound can be shown by adding a reagent, warming the mixture and testing the gas evolved.

What is the reagent and the result of the test for the gas evolved?

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	Reagent	Result of test for gas
☒ A	HCl(aq)	damp blue litmus paper turns red
☒ B	HCl(aq)	damp red litmus paper turns blue
☒ C	NaOH(aq)	damp blue litmus paper turns red
☒ D	NaOH(aq)	damp red litmus paper turns blue

(Total for question = 1 mark)

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Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What produces the colour in a flame test?

	Energy change of electrons	Movement of electrons
☒ A	energy absorbed	from ground state to excited state
☒ B	energy emitted	from ground state to excited state
☒ C	energy absorbed	from excited state to ground state
☒ D	energy emitted	from excited state to ground state

(Total for question = 1 mark)

Q30.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A titration is carried out by adding dilute sulfuric acid from a burette to aqueous sodium hydroxide in a conical flask. The indicator is methyl orange.

(a) What is the colour change of the indicator at the end-point of this titration?

(1)

☒ **A** red to orange
☒ **B** red to yellow
☒ **C** yellow to orange
☒ **D** yellow to red

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(b) A student carried out the first titration and did not notice that there was an air bubble between the tap and the tip of the burette.

During the titration, the air bubble filled with acid.
The student then carried out two accurate titrations in which there was no air bubble in the burette.
There were no other errors in these titrations.
Which of these could be the three titres for this student?

(1)

	First titre / cm^3	Second titre / cm^3	Third titre / cm^3
☒ **A**	22.65	22.65	22.65
☒ **B**	22.65	22.35	23.25
☒ **C**	22.80	22.35	22.40
☒ **D**	22.80	23.25	23.30

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(c) 25.0 cm^3 of 0.100 mol dm^{-3} aqueous sodium hydroxide required a mean titre of 18.70 cm^3 of sulfuric acid.

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

What is the concentration, in mol dm^{-3} , of the sulfuric acid?

(1)

☒ **A** 0.0374
☒ **B** 0.0668
☒ **C** 0.134
☒ **D** 0.267

(Total for question = 3 marks)

Q31.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A titre of 13.25 cm^3 was obtained using a 50 cm^3 burette.

What is the percentage uncertainty in the titre?

[Each reading of the burette has an uncertainty of ± 0.05 cm^3]

☒ **A** ± 0.38 %
☒ **B** ± 0.75 %
☒ **C** ± 1.5 %
☒ **D** ± 7.5 %

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(Total for question = 1 mark)

Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When solid sodium chloride is placed in a Bunsen flame, a persistent yellow colour is observed.

The yellow colour is due to

☐ **A** electrons being excited to higher energy levels and emitting yellow light on returning to the ground state

☐ **B** electrons being excited to higher energy levels and emitting blue-violet light on returning to the ground state

☐ **C** electrons being excited to higher energy levels and absorbing yellow light

☐ **D** electrons being excited to higher energy levels and absorbing blue-violet light

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(Total for question = 1 mark)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A piece of sodium weighing 1.15 g reacts completely with distilled water and the solution formed is made up to 250.0 cm^3.

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

What is the concentration of the sodium hydroxide solution?

☐ **A** 0.025 mol dm^{-3}

☐ **B** 0.050 mol dm^{-3}

☐ **C** 0.100 mol dm^{-3}

☐ **D** 0.200 mol dm^{-3}

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(Total for question = 1 mark)

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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When lithium reacts with excess oxygen, only lithium oxide is formed.
When sodium reacts with excess oxygen, sodium oxide and sodium peroxide are both formed.

What is the main reason why sodium peroxide is formed?

- ☒ **A** sodium is more reactive than lithium
- ☒ **B** sodium has a lower first ionisation energy than lithium
- ☒ **C** the sodium ion has a larger ionic radius than the lithium ion
- ☒ **D** the sodium ion has a higher charge density than the lithium ion

(Total for question = 1 mark)

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Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Compound **Q** produces

- a red colour in a flame test
- a white precipitate when aqueous potassium sulfate is added to a solution of the compound.

What is compound **Q**?

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- ☒ **A** LiCl
- ☒ **B** $NaNO_3$
- ☒ **C** $Sr(NO_3)_2$
- ☒ **D** $BaCl_2$

(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction produces more than one product?

☐ **A** $Mg + O_2 \rightarrow$

☐ **B** $Ca + Cl_2 \rightarrow$

☐ **C** $Sr + H_2O \rightarrow$

☐ **D** $BaO + H_2O \rightarrow$

(Total for question = 1 mark)

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Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Magnesium nitrate decomposes on heating.

$$2Mg(NO_3)_2(s) \rightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$$

A sample of magnesium nitrate decomposed completely.

The total volume of gas formed was 1200 cm^3, measured at room temperature and pressure (r.t.p.).

What mass of magnesium nitrate was used?

[M_r $Mg(NO_3)_2$ = 148.3 Molar volume of gas at r.t.p. = 24 000 cm^3 mol^{-1}]

☐ **A** 1.48 g

☐ **B** 2.97 g

☐ **C** 7.42 g

☐ **D** 14.8 g

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(Total for question = 1 mark)

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Q38.

* Discuss some aspects of the thermal stability of the anhydrous nitrates of the elements in Groups 1 and 2 of the Periodic Table.

In your answer you should

- explain the trend in thermal stability of the **Group 2** nitrates
- describe any differences in the products of thermal decomposition of the **Group 1** nitrates
- give equations for the thermal decomposition of sodium nitrate and of magnesium nitrate.

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(Total for question = 6 marks)

Q39.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which property **decreases** as Group 2 is descended?

- ☒ **A** atomic radius
- ☒ **B** reactivity of the elements
- ☒ **C** solubility of the sulfates
- ☒ **D** thermal stability of the nitrates

(Total for question = 1 mark)

Q40.

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Which statement is **not** explained by hydrogen bonding?

(1)

- ☒ **A** all Group 1 hydroxides are soluble in water
- ☒ **B** many simple alcohols are soluble in water
- ☒ **C** the density of ice is less than the density of liquid water at 0 °C
- ☒ **D** the melting temperature of water is abnormally high

(Total for question = 1 mark)

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Q41.

Compound X gives a red flame test colour and a white precipitate on addition of dilute hydrochloric acid followed by barium chloride solution. Which compound is X?

(1)

- ☒ **A** calcium chloride
- ☒ **B** lithium sulfate
- ☒ **C** potassium sulfate
- ☒ **D** strontium chloride

(Total for question = 1 mark)

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Q42.

Which process explains the flame colour produced by the compounds of Group 1 elements?

(1)

- ☒ **A** absorption of visible light energy as electrons are promoted to higher energy levels
- ☒ **B** absorption of visible light energy as electrons are removed from gaseous atoms
- ☒ **C** emission of visible light energy as electrons return to lower energy levels
- ☒ **D** emission of visible light energy as electrons are added to gaseous ions

(Total for question = 1 mark)

Q43.

Ethanedioic acid is a solid diprotic acid. A student used ethanedioic acid in a titration to find the concentration of a potassium hydroxide solution.

The equation for the reaction is:

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$$2KOH + (COOH)_2 \rightarrow (COOK)_2 + 2H_2O$$

(a) Calculate the mass of ethanedioic acid that should be used to make 1000 cm^3 of a 0.0500 mol dm^{-3} solution in water.

Give your answer to an appropriate number of significant figures.
[Molar mass of ethanedioic acid = 90.0 g mol^{-1}].

(2)

(b) A student decided to check to see if phenolphthalein was a suitable indicator for this titration. The student measured 400 cm^3 of the 0.0500 mol dm^{-3} ethanedioic acid into a beaker and added a few drops

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of phenolphthalein indicator.

Calculate the minimum mass of solid potassium hydroxide that should be added to produce a colour change.

(2)

*(c) A student used a 0.0500 mol dm^{-3} solution of ethanedioic acid to find an accurate concentration of a potassium hydroxide solution which was known to have an approximate concentration of 0.1 mol dm^{-3}.

Describe a procedure to obtain reliable titration results using standard laboratory equipment.

(6)

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(Total for question = 10 marks)

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Q44.

A colourless solid, Q, was warmed with sodium hydroxide solution. A gas was evolved which turned damp red litmus paper blue. What is solid Q?

(1)

- ☒ **A** $NaNO_3$
- ☒ **B** NH_4Cl
- ☒ **C** NaCl
- ☒ **D** $Ca(NO_3)_2$

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(Total for question = 1 mark)

Q45.

A student made the following statements about trends going **down** Group 2.
Which statement is correct?

(1)

- ☒ **A** the thermal stability of the nitrates decreases
- ☒ **B** the thermal stability of the carbonates decreases
- ☒ **C** the solubility of hydroxides increases
- ☒ **D** the solubility of sulfates increases

(Total for question = 1 mark)

Q46.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A compound, which is a white solid, gives a red flame test and produces a colourless, acidic gas when heated.

What is the identity of the compound?

- ☒ **A** calcium carbonate
- ☒ **B** calcium nitrate
- ☒ **C** sodium carbonate
- ☒ **D** strontium nitrate

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(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which statement about Group 2 elements and their compounds is **true**?

- ☒ **A** calcium hydroxide is less soluble than magnesium hydroxide in water
- ☒ **B** all Group 2 metals give a characteristic flame test colour
- ☒ **C** magnesium carbonate does **not** thermally decompose when heated
- ☒ **D** the first ionisation energy of magnesium is greater than that of calcium

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(Total for question = 1 mark)

Q48.

This question is about lithium and some of its compounds.

Lithium has an atomic number of 3 and is the least dense of all the solid elements.

(a) Lithium hydroxide solution can be made by reacting lithium with water.

(i) Write an equation for the reaction that occurs.
Include state symbols in your answer.

(2)

(ii) Give **two** observations you would make during this reaction.

(2)

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(b) Lithium metal is stored under oil.

When a piece of lithium metal is removed from the oil, the surface quickly tarnishes. One compound formed is lithium nitride.
State how lithium nitride is formed, including the formula of lithium nitride.

(2)

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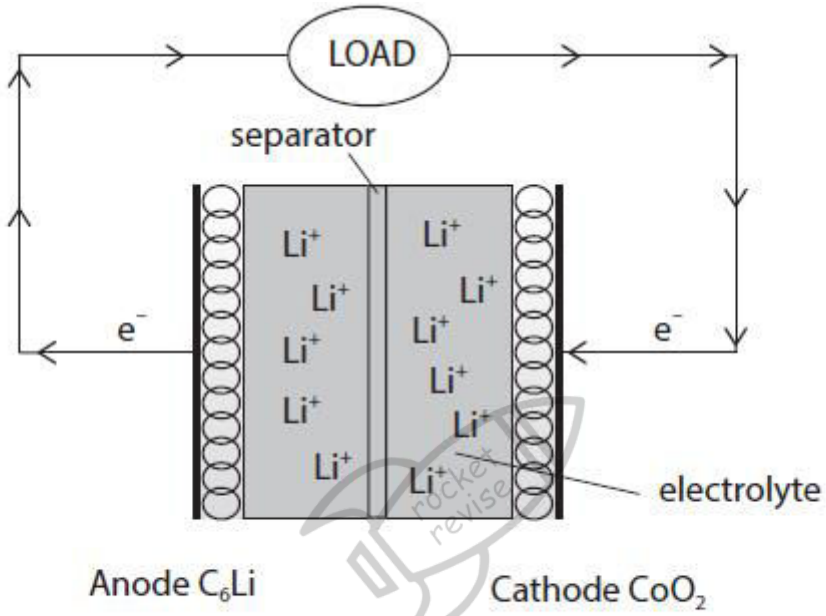
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(c) Mobile phones rely on rechargeable lithium batteries.

The cell of one type of battery consists of an anode containing a compound of lithium and carbon, C_6Li, and a cathode containing cobalt(IV) oxide, CoO_2.
The two electrodes are kept apart by a separator.
When the cell is in use and current flows, at the anode the Li atoms lose electrons and form Li^+ ions.
The ions travel through the separator and the electrons travel through the external circuit to the cathode.
At the cathode, the cobalt(IV) oxide combines with the Li^+ ions and electrons and forms $LiCoO_2$.



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Reactions: $C_6Li \rightarrow C_6 + Li^+ + e^-$ $\quad CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$

(i) Give the overall equation when the cell is being used.
State symbols are not required.

(1)

(ii) Describe, with reference to oxidation numbers, what happens to Co when the cell is in use.

(2)

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(iii) Suggest **one** property the separator must have to ensure the lithium battery is viable.

(1)

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(d) Lithium is often added to aluminium to make aluminium-lithium alloys.

These alloys have a variety of uses, including aircraft manufacture.
One alloy contains only aluminium and lithium.

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It contains 2.45% of Li by mass.

(i) Calculate the molar ratio of Al : Li in this alloy.

(3)

(ii) Suggest a reason why this alloy is better than pure aluminium for aircraft manufacture.

(1)

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*(e) Compounds of lithium (Group 1) and compounds of magnesium (Group 2) have similar properties.

Explain why:

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- the ionic radii of Li^+ and Mg^{2+} are similar
- the thermal decomposition of lithium nitrate is similar to that of magnesium nitrate but different from the thermal decomposition of sodium nitrate.

Include relevant chemical equations in your answer.

(6)

(Total for question = 20 marks)

Q49.

This question is about magnesium sulfate, a white ionic solid that is very soluble in water.

Hydrated magnesium sulfates have the general formula of $MgSO_4 \cdot xH_2O$, where x is the number of water molecules of crystallisation.

(a) A sample of hydrated magnesium sulfate was heated until all the water had been removed from the crystals.

The mass of the sample decreased from 6.92 g to 6.04 g.

[M_r $MgSO_4$ = 120.4]

(i) State how you could ensure that all the water had been removed from the crystals.

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(1)

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(ii) Calculate the number of water molecules of crystallisation, x, in the formula of this sample of hydrated magnesium sulfate.

(3)

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(b) The most common form of hydrated magnesium sulfate is $MgSO_4 \cdot 7H_2O$.

A student carried out two experiments to determine the enthalpy change when anhydrous magnesium sulfate forms $MgSO_4 \cdot 7H_2O$.

$$MgSO_4(s) + 7H_2O(l) \rightarrow MgSO_4 \cdot 7H_2O(s)$$

(i) In the first experiment, the student determined the enthalpy change when dissolving anhydrous magnesium sulfate.

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$$MgSO_4(s) + (aq) \rightarrow MgSO_4(aq)$$

100.0 g of distilled water was placed in a polystyrene cup and the temperature recorded.
0.0628 mol of anhydrous magnesium sulfate was added to the distilled water, the mixture stirred and the maximum temperature recorded.

Results

Starting temperature of distilled water / °C	16.6
Maximum temperature of solution / °C	29.4

Calculate the enthalpy change for this reaction.
Give your answer to an appropriate number of significant figures and include a sign and units.
[Assume: Specific heat capacity of the solution is 4.18 J g^{-1} $°C^{-1}$ Density of the solution is 1.00 g cm^{-3}]

(4)

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(ii) In the second experiment, the student determined the enthalpy change when dissolving hydrated magnesium sulfate.

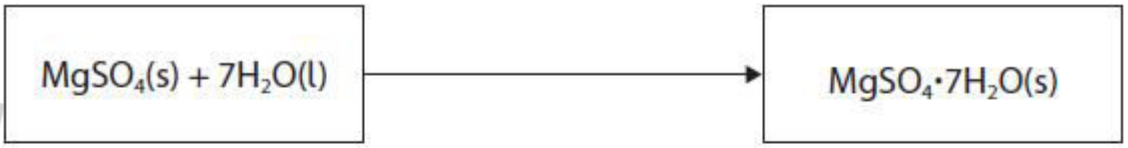
$$MgSO_4 \cdot 7H_2O(s) + (aq) \rightarrow MgSO_4(aq) \qquad \Delta H = +15.8\,kJ\,mol^{-1}$$

Complete the Hess cycle.

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(2)





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(iii) Calculate the enthalpy change when anhydrous magnesium sulfate forms hydrated magnesium sulfate, using the completed Hess cycle and your answer in (b)(i).

(2)

(c) Magnesium sulfate is very soluble in water.

(i) Draw a labelled diagram showing how both the magnesium ion and the sulfate ion interact with water molecules.

(2)

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(ii) Barium ions are toxic in aqueous solution. If a solution containing barium ions enters the body, barium poisoning occurs.

Describe how drinking magnesium sulfate solution can reduce the extent of the poisoning.
Include an ionic equation with state symbols in your answer.

(2)

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(Total for question = 16 marks)

Q50.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Steam is passed over heated magnesium.

Which reaction occurs most readily?

- ☐ **A** $Mg + H_2O \rightarrow MgO + H_2$
- ☐ **B** $Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$
- ☐ **C** $2Mg + H_2O \rightarrow Mg_2O + H_2$
- ☐ **D** $2Mg + 2H_2O \rightarrow 2MgOH + H_2$

(Total for question = 1 mark)

Q51.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In a titration, a conical flask containing 10.00 cm^3 of 1.00 mol dm^{-3} sulfuric acid is fully neutralised by 25.00 cm^3 of 0.80 mol dm^{-3} potassium hydroxide.

(a) The titration is carried out using methyl orange as an indicator.

What would be the colour change at the end-point of the titration?

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(1)

- ☐ **A** orange to red
- ☐ **B** red to orange
- ☐ **C** orange to yellow
- ☐ **D** yellow to orange

(b) What is the concentration, in mol dm^{-3}, of the potassium sulfate solution produced by this reaction at the end-point, to 2 significant figures?

(1)

- ☐ **A** 0.010
- ☐ **B** 0.020
- ☐ **C** 0.29
- ☐ **D** 0.57

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(c) Information about the uncertainties in each reading used to measure the volumes of the solutions in this titration is shown.

Solution	Equipment used	Uncertainty / cm^3
sulfuric acid	pipette	±0.04
potassium hydroxide	burette	±0.05

Which statement is true?

(1)

☐ **A** the percentage uncertainty of the volume of sulfuric acid is more than the percentage uncertainty of the volume of potassium hydroxide

☐ **B** the percentage uncertainty of the volume of sulfuric acid is 0.8 %

☐ **C** the total percentage uncertainty is 0.8 %

☐ **D** the percentage uncertainty of the volume of potassium hydroxide solution is 0.2 %

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(Total for question = 3 marks)

Q52.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A 20 g sample of iron(III) sulfate, $Fe_2(SO_4)_3$, was dissolved in water.

How many sulfate ions are in the solution formed?

[Assume M_r $Fe_2(SO_4)_3$ = 400
Avogadro constant, L = 6.0 × 10^{23} mol^{-1}]

☐ **A** 3 × 10^{22}

☐ **B** 9 × 10^{22}

☐ **C** 1.2 × 10^{23}

☐ **D** 1.5 × 10^{23}

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(Total for question = 1 mark)

Q53.

This question is about the element calcium and some of its compounds.

(a) Give the complete electronic configuration, using s, p, d notation, of a calcium atom in the ground

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state.

(1)

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(b) Calcium reacts with chlorine to form calcium chloride.

$$Ca(s) + Cl_2(g) \rightarrow CaCl_2(s)$$

(i) Explain, in terms of oxidation numbers, why this is a redox reaction.

(2)

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(ii) Complete the table to indicate the type(s) of bonding and structure in the species involved in this reaction.

(3)

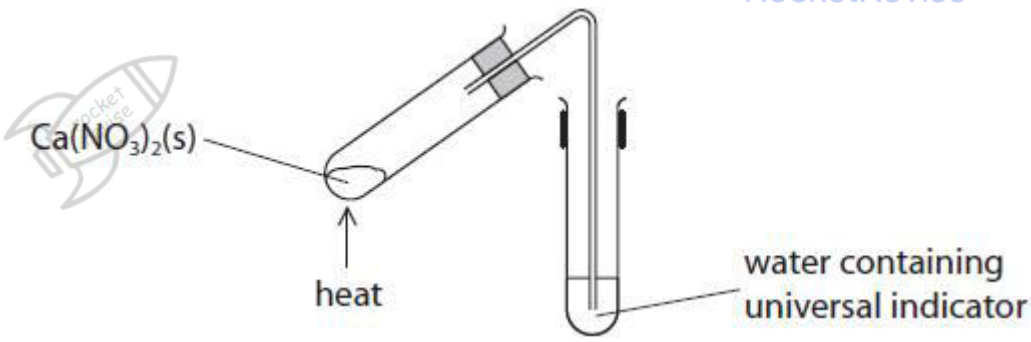
Species	Bonding	Structure
Ca(s)		
	covalent	
		giant

(iii) State the expected flame colour when $CaCl_2$ is used in a flame test.

(1)

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(c) A sample of solid anhydrous calcium nitrate, $Ca(NO_3)_2(s)$, is heated strongly for several minutes in the apparatus shown.



(i) State **two** observations that would be made.

(2)

1 .....

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(ii) Explain why the observations from (c)(i) would be made more quickly if the experiment was repeated with solid anhydrous magnesium nitrate, $Mg(NO_3)_2(s)$.

(3)

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*(d) Samples of calcium and magnesium are added to separate test tubes of cold water containing a few drops of universal indicator.

State and explain the differences in the observations made in each experiment.
Include an equation in your answer.

(6)

(Total for question = 18 marks)

Q54.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Group 2 compounds give different colours in a flame test.

The **best** explanation is that the cations have different

- ☒ **A** gaps between electronic energy levels
- ☒ **B** ionic radii
- ☒ **C** ionisation energies
- ☒ **D** numbers of electrons

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

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Q55.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which test reagent could be used to identify an aqueous solution containing hydrogencarbonate ions, HCO^-_3 ?

- ☒ **A** sodium carbonate
- ☒ **B** sodium hydroxide
- ☒ **C** nitric acid
- ☒ **D** ammonia

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(Total for question = 1 mark)

Q56.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which statement explains the origin of a flame test colour?

- ☒ **A** electrons absorb energy as they jump to a higher energy level
- ☒ **B** electrons emit energy as they jump to a higher energy level
- ☒ **C** electrons absorb energy as they drop to a lower energy level
- ☒ **D** electrons emit energy as they drop to a lower energy level

(Total for question = 1 mark)

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Q57.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What are the products formed when lithium nitrate and calcium nitrate are heated separately?

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	Lithium nitrate	Calcium nitrate
☒ **A**	lithium nitrite and oxygen	calcium nitrite and oxygen
☒ **B**	lithium nitrite and oxygen	calcium oxide, nitrogen dioxide and oxygen
☒ **C**	lithium oxide, nitrogen dioxide and oxygen	calcium nitrite and oxygen
☒ **D**	lithium oxide, nitrogen dioxide and oxygen	calcium oxide, nitrogen dioxide and oxygen

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(Total for question = 1 mark)

Q58.

A sample of 25.0 cm^3 of 0.205 mol dm^{-3} sulfuric acid was titrated with sodium hydroxide solution. The titre obtained was 14.60 cm^3 .

(a) What is the concentration of the sodium hydroxide solution?

(1)

☒ **A** 0.176 mol dm^{-3}
☒ **B** 0.239 mol dm^{-3}
☒ **C** 0.351 mol dm^{-3}
☒ **D** 0.702 mol dm^{-3}

(b) What is the percentage error in the titre?

[The burette is accurate to 0.05 cm^3 per reading]

(1)

☒ **A** 0.10%
☒ **B** 0.34%
☒ **C** 0.40%
☒ **D** 0.68%

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(Total for question = 2 marks)

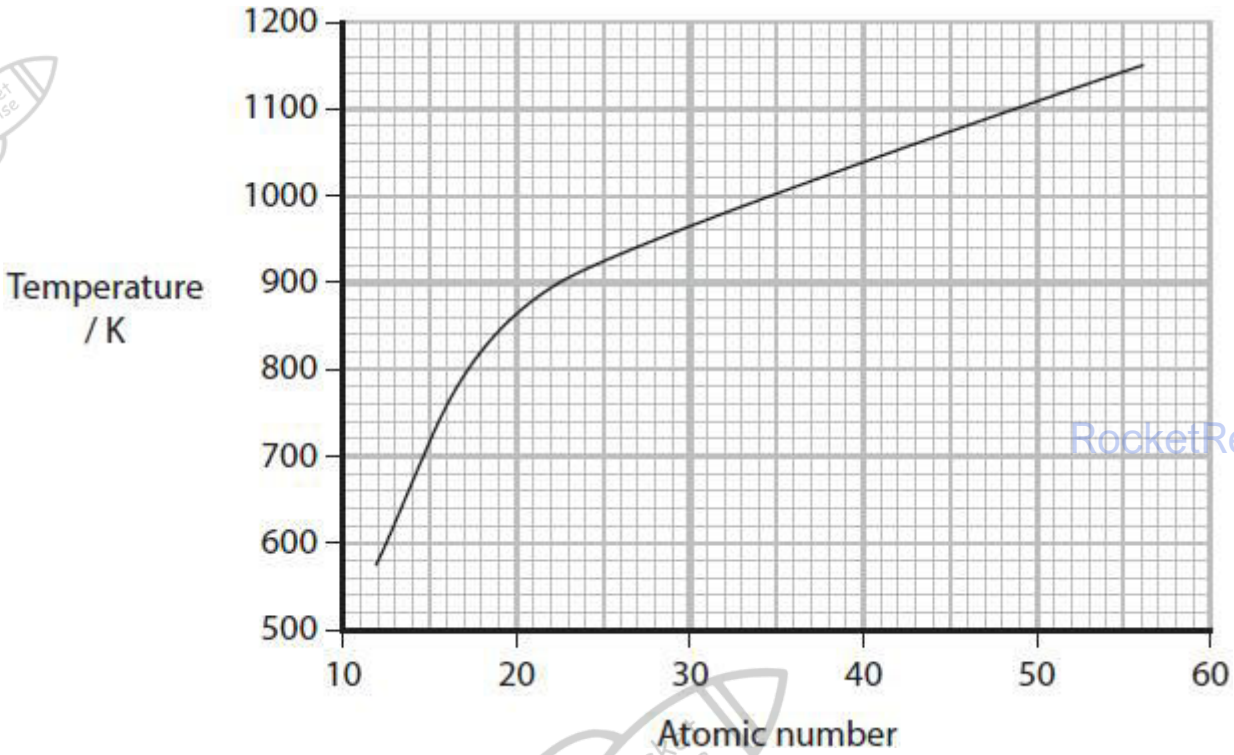
Q59.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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A student sketched a graph of the temperature (K) of the decomposition of some of the Group 2 nitrates against the metal's atomic number.



Use the graph to estimate the temperature, in °C, at which calcium nitrate decomposes.

- ☒ **A** 587°C
- ☒ **B** 767°C
- ☒ **C** 860°C
- ☒ **D** 1040°C

(Total for question = 1 mark)

Q60.

Which order is correct for the solubilities of Group 2 sulfates?

- ☒ **A** barium > strontium > magnesium
- ☒ **B** calcium > strontium > magnesium
- ☒ **C** magnesium > barium > calcium
- ☒ **D** calcium > strontium > barium

(Total for question = 1 mark)

Q61.

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20 cm^3 of a saturated solution of barium nitrate at 80 °C was cooled to 20 °C. 9.32 g of crystals were precipitated.

The solubility of barium nitrate is 3.74 g per 100 g water at 20 °C.

What is the solubility of barium nitrate at 80 °C in g per 100 g water?

- **A** 13.1
- **B** 41.0
- **C** 46.6
- **D** 50.3

(Total for question = 1 mark)

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Q62.

What is the flame colour produced by strontium ions?

- **A** green
- **B** red
- **C** white
- **D** yellow

(Total for question = 1 mark)

Q63.

Magnesium nitrate decomposes on heating according to the equation shown.

$$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$$

One mole of magnesium nitrate is completely decomposed.

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(a) What is the maximum volume of gas formed at 600 K and 101 000 Pa?

[$pV = nRT$ $R = 8.31$ J mol^{-1} K^{-1}]

(1)

- **A** 0.025 m^3
- **B** 0.099 m^3
- **C** 0.123 m^3
- **D** 0.247 m^3

(b) What is the percentage yield if 25.0 g of magnesium oxide is produced?

[M_r MgO = 40.3]

(1)

- **A** 13.6 %
- **B** 16.9 %

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- **C** 31.0 %
- **D** 62.0 %

(Total for question = 2 marks)

Q64.

What is the concentration, in mol dm^{-3} , when 64.5 g of sodium nitrate is dissolved in water to form 750 cm^3 of solution?

[M_r $NaNO_3$ = 85.0]

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- **A** 0.086
- **B** 0.113
- **C** 0.759
- **D** 1.012

(Total for question = 1 mark)

Q65.

A volume of 10.0 cm^3 of magnesium chloride solution was added to an excess of sodium hydroxide solution, forming a precipitate of magnesium hydroxide. After filtering and drying, 0.398 g of magnesium hydroxide was obtained.

What was the concentration of the magnesium chloride solution?

[M_r $Mg(OH)_2$ = 58.3]

- **A** 0.00683 mol dm^{-3}
- **B** 0.0137 mol dm^{-3}
- **C** 0.683 mol dm^{-3}
- **D** 1.37 mol dm^{-3}

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(Total for question = 1 mark)

Q66.

Calcium carbonate can be heated to produce calcium oxide.

$$CaCO_3 \rightarrow CaO + CO_2$$

What is the atom economy, by mass, for the formation of calcium oxide?

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- ☐ **A** 44.0%
- ☐ **B** 56.0%
- ☐ **C** 78.6%
- ☐ **D** 100%

(Total for question = 1 mark)

Q67.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

25 cm^3 of sodium hydroxide solution, of concentration 2.00 mol dm^{-3}, was added to 30 cm^3 of ethanoic acid solution, of concentration 1.00 mol dm^{-3}, in a polystyrene cup.
The mixture was stirred, causing the temperature to increase by 6.5 °C.

(a) What is the expression for the enthalpy change of neutralisation of sodium hydroxide and ethanoic acid, in J mol^{-1}?

[Assume: the specific heat capacity of the mixture = 4.18 J g^{-1} $°C^{-1}$
the density of the mixture = 1.00 g cm^{-3}]

(1)

- ☐ **A** $-(25 \times 4.18 \times 6.5) \div 0.05$
- ☐ **B** $-(30 \times 4.18 \times 6.5) \div 0.03$
- ☐ **C** $-(55 \times 4.18 \times 6.5) \div 0.05$
- ☐ **D** $-(55 \times 4.18 \times 6.5) \div 0.03$

(b) The measurement uncertainty in the thermometer used in the experiment is ±0.5 °C for each reading.

What is the percentage uncertainty in the temperature change of 6.5 °C?

(1)

- ☐ **A** 3.8%
- ☐ **B** 7.7%
- ☐ **C** 15.4%
- ☐ **D** 30.8%

(Total for question = 2 marks)

Q68.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which electron movement causes the yellow colour produced in a sodium chloride flame test?

- ☐ **A** addition of an electron to a sodium ion
- ☐ **B** an electron moving from a 3p to a 3s orbital

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☒ **C** an electron moving from a 3s to a 3p orbital
☒ **D** removal of an electron from a sodium atom

(Total for question = 1 mark)

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